460 [The seduction began in 1952,] Ibid.

462 [One day in late autumn of 1964, . . . ] INT-Penrose; Penrose (1989).

462 [“My conversation with Robinson . . . crossing the street.”] Penrose (1989).

462 [a short article for. . . *Physical Review Letters*,] Penrose (1965).

465 [*global methods*.] The global methods were codified in a classic book by Hawking and Ellis (1973).

465 [Hawking and Penrose in 1970 proved . . . big crunch.] Hawking and Penrose (1970).

466 [Lifshitz, though Jewish, . . . 1976.] From my private discussions with Lifshitz in the 1970s.

466 [Khalatnikov had two strikes against him;. . . come to London.] Letter from Khalatnikov to me, 18 June 1990.

466 [As he spoke in the packed London lecture hall, . . . Penrose, they asserted, was probably wrong.] From my own memory of the meeting and its aftermath.

468 [“Please, . . . submit it to *Physical Review Letters*,”] Khalatnikov and Lifshitz (1970). See also Belinsky, Khalatnikov, and Lifshitz (1970, 1982).

468 [I carried the manuscript . . . published.] Ibid.

470 [Lev Davidovich Landau . . . great physics discoveries.] INT-Lifshitz, Livanova (1980).

471 [Curiously, topological techniques . . . Pimenov.] I learned this from Penrose.

471 [In 1950–59, Aleksandrov . . . that cannot.] Aleksandrov (1955, 1959).

471 [picked up and pushed further by Pimenov,] Pimenov (1968).

473 [(due to Khalatnikov and Lifshitz) . . . “unstable against small perturbations.”] Lifshitz and Khalatnikov (1960, 1963).

473 [The Reissner–Nordström . . . large universe] e.g., Novikov (1966).

473–474 [it is unstable . . . many different physicists.] In technical language, it is the *inner Cauchy horizon* of the Reissner–Nordström solution that is unstable. The conjecture is in Penrose (1968); the proofs are in Chandrasekhar and Hartle (1982) and earlier references cited therein.

474 [Belinsky, Khalatnikov, and Lifshitz . . . (This is the kind . . . holes.)] Belinsky, Khalatnikov, and Lifshitz (1970, 1982).

474 [Charles Misner . . . *mixmaster oscillation*] Misner (1969).

476 [Just when does quantum gravity take over, . . . or less.] This was first deduced by Wheeler (1960), building on his own earlier ideas of vacuum fluctuations of the geometry of spacetime (Wheeler, 1955, 1957).

476 Footnote 2: The *Planck–Wheeler time* was introduced and its physical significance deduced by Wheeler (1955, 1957).

476–477 [Quantum gravity then radically changes . . . random, probabilistic froth,] This was first suggested by Wheeler (1960), and has been made more quantitative since via what is now called the “Wheeler–DeWitt equation.” See, e.g., the discussion in Hawking (1987).

477 [John Wheeler,. . . *quantum foam*] Wheeler (1957, 1960).

479 [Clear answers . . . DeWitt.] See, e.g., Hawking (1987, 1988).

479 [The tidal forces . . . and gradually disappear.] Doroshkevich and Novikov (1978) showed that the singularity ages; Poisson and Israel (1990) and Ori (1991) deduced the details of the aging in idealized models; and Ori (1992) has tentatively shown that these models are good guides to the behavior of singularities in real black holes.

481 [Some implosions,. . . might actually create naked singularities.] For details of these simulations see Shapiro and Teukolsky (1991).